MILITARY SPECIFICATION

PUMP, VACUUM, POSITIVE DISPLACEMENT, AIR ROTARY VANE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and test of types 1 and 2 rotary vane air pumps, hereinafter referred to as a pump.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-665	-	Box, Paperboard, Metal Edged and Components.

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-S-901	-	Shock Tests, H. I. (High-impact), Shipboard Machinery, Equipment and Systems Requirements for (Navy).
MIL-Q-9858	-	Quality Program Requirements
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-P-26514	-	Polyurethane Foam, Rigid or Plastic, for Packaging.
MIL-G-81322	-	Grease, Aircraft, General Purpose Wide Temperature Range.
MIL-B-81793	-	Bearings, Ball, Precision, for Instruments and Rotating Components.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-781	-	Reliability Test Exponential Distribution.

(Copies of specifications, standards, and publications required by the manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:15Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.1.1 First article. When specified, a sample shall be subjected to the first article inspection (see 4.3 and 6.3.)

3.3 Materials. The materials used shall be as specified. (see 3.1)

3.4 Design and construction. The pump shall be fabricated and assembled in accordance with the drawings, parts list, and other documentation as specified (see 3.1). The design of the equipment shall be such that performance of the pump shall not depend on the individual selection of parts for characteristics which are not specified or to tolerances closer than those specified in the general or detail specifications.

3.5 Reliability. The pump shall have a lower test mean-time-between-failure (MTBF) (θ_1) of 3,000 hours and an upper test MTBF (θ_0) of 9,000 hours while operating in an air ambient of 65°C±2.

3.6 Maintainability. The pump shall be capable of being disassembled without destruction and work parts replaced to restore required performance at Depot level. The pump shall require no periodic maintainance or lubrication for the design MTBF.

3.7 Performance Characteristics.

3.7.1 General. Unless otherwise specified, tolerances for required values shall be ±5 percent and Standard Temperature and Pressure (STP) shall be 25°C±5 and 30±.5 inches mercury respectively.

3.7.2 Inlet vacuum. The inlet vacuum that the pump is capable of shall be not less than 15 inches of mercury with a pump speed of 1420 revolutions per minute (RPM) under STP.

3.7.3 Outlet pressure. The outlet pressure shall be not less than 15 pounds per square inch gage (psig) with pump speed of 1420 RPM under STP conditions.

3.7.4 Flow rate. The air volume flow rate shall be not less than 3.5 cubic feet per minute (cfm) under STP conditions and a speed of 1420 RPM. The air flow rate shall be not less than 2.1 cfm with an inlet vacuum of 8 inches of mercury and an outlet pressure of 2 psig with external conditions STP and the pump rotating at 1420 RPM.

3.7.5 Efficiency. The power requirements for operating the pump using a 1/3-horse power century motor with the pump rotating at 1420 RPM and the inlet vacuum of 8 inches of mercury and the outlet pressure of 2 psig, shall be not greater than 600 watts.

3.7.6 Temperature. The temperature rise as measured on pump body shall be not greater than 50°C above actual temperature of the test room which shall be maintained within STP. The pump shall be run under the conditions of 3.7.5 until the temperature rise stablizes at less than 1°C rise per 10 minutes before making measurements.

3.7.7 Pressure. The pump shall be capable of running for 10 minutes under an internal pressure of 150 psig. The leakage rate of the pump under these conditions shall not exceed one cubic foot per hour (cfh).

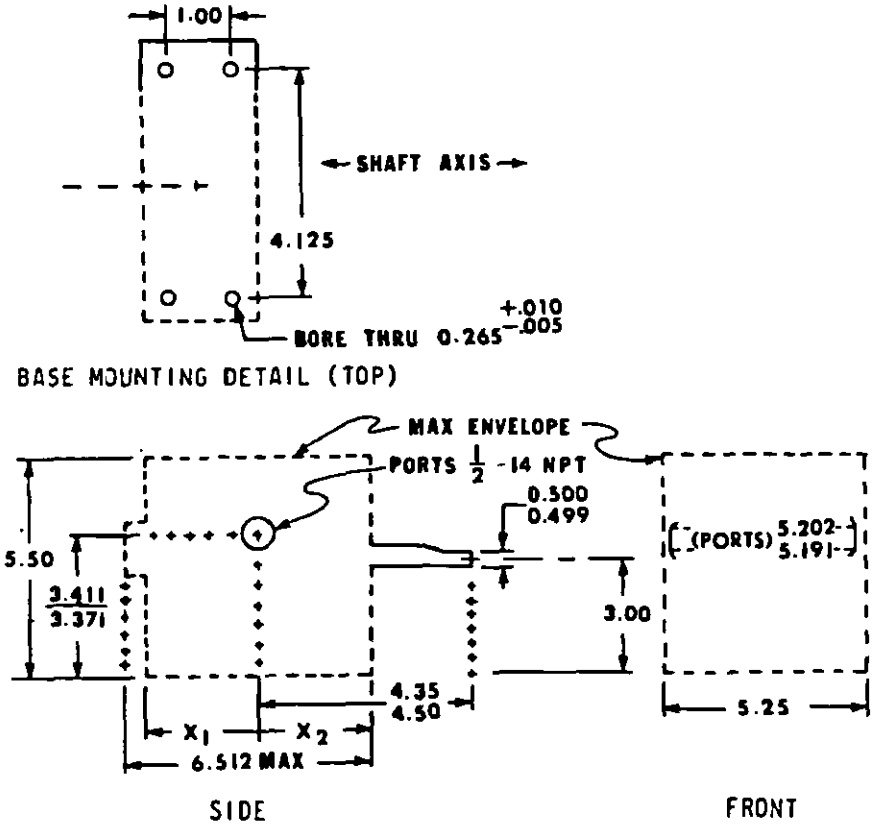
Source: https://assist.dla.mil -- Downloaded: 202610-20T19:15Z
Check the source to verify that this is the current version before use.

3.7.8 Environment. The pump shall be capable of operating without degradation in performance in an ambient temperature range of 0 to 65°C and humidity conditions of 95 percent relative humidity at 65°C, while pumping the ambient air.

3.7.9 Operating stress. The pump shall be capable of operating with the intake sealed for a period of four hours without damage.

3.8 Details of components. The bearings shall be in accordance with MIL-B-81793 and permanently lubricated with MIL-G-81322 high temperature grease. The amount of grease fill shall be 25 percent of the void volume.

3.9 Dimensional requirements. The pump shall be in accordance with Figure 1 and as specified (see 3.1).



NOTE: All dimensions are .01 inches, unless specified.

FIGURE 1. Base mounting detail and maximum envelope size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:15Z
Check the source to verify that this is the current version before use.

3.10 Weight. The weight shall be as specified (see 3.1).

3.11 Color and finish. The color and finish shall be as specified (see 3.1).

3.12 Markings.

3.12.1 General. Markings shall be applied by stamping, stenciling or rubber stamping with smudge-proof ink covered with a coat or clear lacquer or shall be silk screened. Decals may be used to show direction or rotation, inlet and outlet ports, and "Warning" information.

3.12.2 Identification and serial number. Each pump shall have the manufacturer's name, national stock number, contract number, this specification number and slash sheet number, and serial number stamped, cast, or attached by metal tag and drive rivets, on the pump body. This serial number shall be made up as follows:

(a) First two numbers, designate the last two numbers of the year of manufacturer.
(b) Next two numbers, the number of the week the unit was manufactured.
(c) Last three numbers, the consecutive number of completions during that week.

EXAMPLE: The 21st pump completed during the week of July 1976 (or 28th week) would be: 76-28-021.

3.13 Government loaned material. The government will loan, upon request by the manufacturer for the period of manufacture only, the specified test equipment as listed in the detail specification. The material shall be managed under the guidelines of MIL-Q-9858 while on loan to the manufacturer.

3.14 Shock. The pump shall comply with grade A, class 1 lightweight type A shock requirements of MIL-S-901 and 4.5.6 herein.

3.15 Workmanship. The highest standard of workmanship shall be utilized in the construction of this pump. Particular attention shall be given to the following:

(a) Burrs and sharp edges shall be removed.
(b) There shall be no rust, corrosion, or foreign matter, including grease, dirt and oil left on the pump surfaces.
(c) The overall appearance and workmanship, where not specified, shall be uniform and reproducible.
(d) All fastners shall be secure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 First article inspection.

4.3.1 Sample size. The sample size shall be five pumps.

4.3.2 Requirements. The tests shall consist of the tests shown in Table I and as specified (see 3.1).

Source: https://assist.dla.mil -- Downloaded: 20260-20T19:15Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

Test and Inspection	Requirement	Method	Sample
Group A			
Visual (external)	3.3, 3.9, 3.10 3.11, 3.12, 3.15	4.5.1	All
Performance	3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5, 3.7.6 3.7.7	4.5.2	All
Group B			
Environment	3.7.8	4.5.4	Four
Stress	3.7.9	4.5.5	Four
Reliability and Life	3.5	4.5.7.1	Four
Group C			
Tear Down (Internal)	3.3, 3.4, 3.6 3.8, 3.9, 3.11 3.15	4.5.3	One
Shock	3.14	4.5.6	One

4.4 Quality conformance inspection.

4.4.1 Requirements. Quality conformance inspection shall consist of the tests shown in Table II and as specified (see 3.1).

TABLE II. Quality conformance inspection.

Test and Inspection	Requirement	Method	Sample
Group A			
Visual	3.3, 3.9, 3.10, 3.11, 3.12, 3.15	4.5.1	MIL-STD-105 Level II
Performance	3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5, 3.7.6, 3.7.7	4.5.2	All

Source: https://assist.dla.mil -- Downloaded: 20260-20T19:15Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance inspection (continued).

Test and Inspection	Requirement	Method	Sample
Group B			
Reliability	3.5	4.5.7.2	15% or 12 units whichever is greater

4.5 Test methods.

4.5.1 Visual examination. An external examination shall be made to ensure that the pump complies with the basic material, workmanship, dimensions and markings as specified (see 3.1).

4.5.2 Performance. After a ten hour period of "run in" the performance tests detailed in Appendix A shall be performed. Pumps not meeting requirements shall be rejected.

4.5.3 Internal visual examination (teardown). After the pump has been checked for performance, it shall then be disassembled and checked for the specified dimension requirements (see 3.1).

4.5.4 Environment. The pump shall be operating during this test and valves installed at the inlet and outlet to maintain 8 inches of mercury inlet vacuum and 2 psig outlet pressure. The pump shall be run for four hours. During the test the ambient temperature shall be maintained at $0^{\circ}C$, $+2 - 0$, for the first hour. For the last three hours the temperature shall be maintained at $65^{\circ}C \pm 2$ and a relative humidity of 95 percent or above. Upon completion of this test, the pump shall pass the performance test as specified in 4.5.2.

4.5.5 Stress test. The pump shall be run for a period of four hours at STP, except the temperature shall be $65^{\circ}C \pm 2$, with the inlet port sealed. Upon completion of this test, the pump shall pass the performance test as specified in 4.5.2.

4.5.6 Shock test. The pump shall have a bracket as shown in Figure 2 and shall be mounted on the high shock test equipment using four 1/4 inch bolts with 20 threads per inch through the holes in the mounting foot. The pump shall not be operating during the shock test. The pump shall be subjected to one and three foot blows only in each of three mutually perpendicular axes (top, shaft end, and side). Upon completion of the shock test, the pump shall pass the performance test as specified in 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 202640-20T19:15Z
Check the source to verify that this is the current version before use.

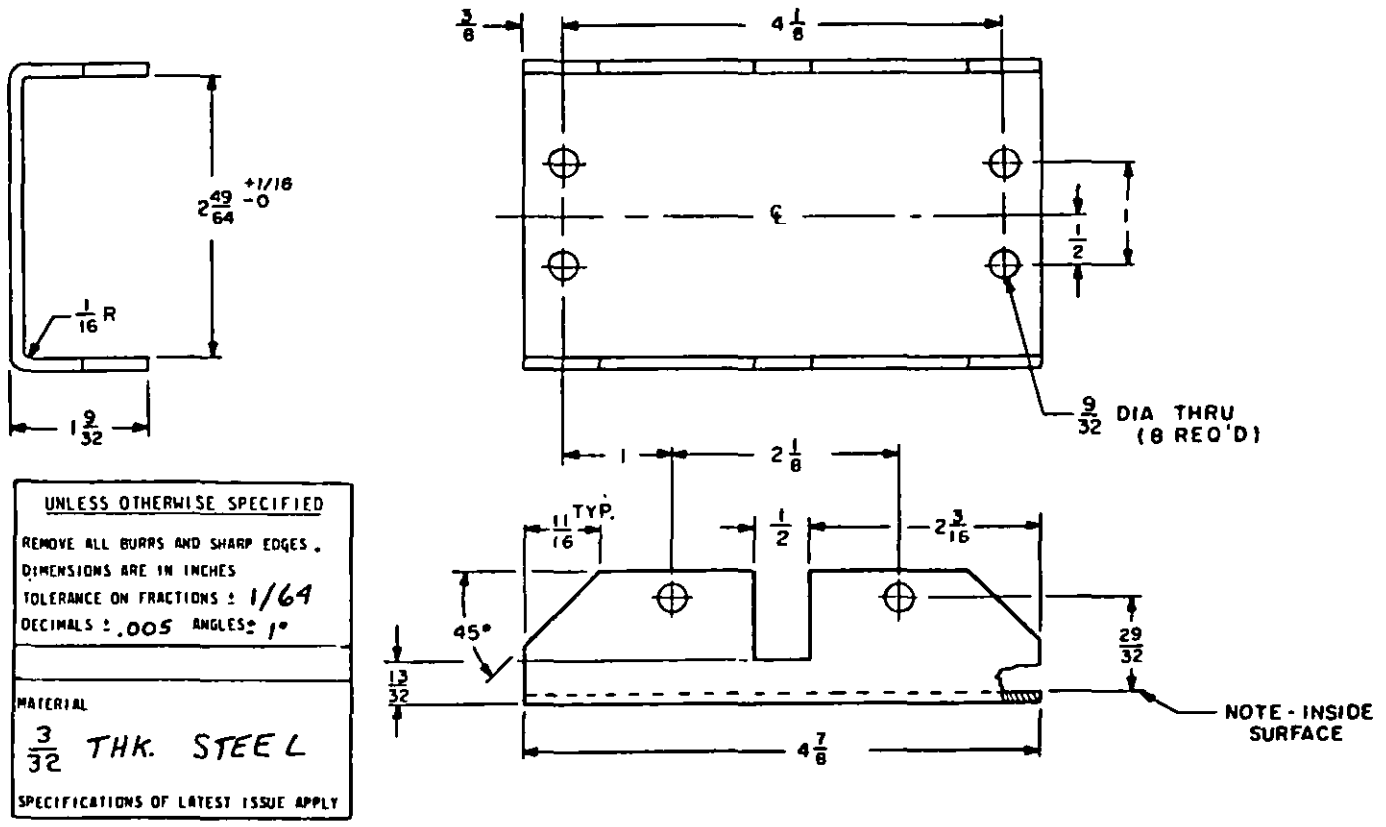


FIGURE 2. Mounting bracket.

4.5.7 Reliability.

4.5.7.1 Reliability first article (preproduction) qualification test. Each pump shall be run in accordance with test plan XXIC of MIL-STD-781 for 500 hours under STP conditions, except the ambient temperature shall be kept at 65 C $^{+2}$ and the pump shall maintain an 8-inch mercury inlet vacuum and a 2 psig outlet pressure. Failures are not permitted. Upon completion of the test, units shall pass the performance test as specified in 4.5.2 and the internal inspection test as specified in 4.5.3, including an inspection of the bearing lubricant condition.

4.5.7.2 Reliability production acceptance test. The sample size shall be 15 percent of the contract quantity or 12 pumps, whichever is greater. The ambient conditions shall be STP, except the temperature shall be kept at 65 $^{+2}$ C. The pump operation (duty cycle) shall be continuous. The total test time and accept/reject criteria shall be in accordance with test plan VC of MIL-STD-781 with a lower test MTBF (θ_1) of 3,000 hours. During the test, pumps shall maintain an 8-inch mercury inlet vacuum and 2 psig outlet pressure. Upon completion of this test, the pump shall pass the performance test as specified in 4.5.2.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct

Source: https://assist.dla.mil -- Downloaded: 20260-20T19:15Z
Check the source to verify that this is the current version before use.

Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation-packaging. Unless otherwise specified in the contract, preservation and packaging shall be level A in accordance with MIL-E-17555.

5.1.1 Preservation. Cleaning shall be in accordance with MIL-P-116.

5.1.2 Packaging. The vacuum pump and motor shall be packaged in a container made in accordance with PPP-B-665, Style A, having glued-in cushioning conforming to MIL-P-26514, Class 2.

5.2 Caps. Units shall have protective plastic caps inserted in threaded inlet and outlet ports.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pump, as specified herein, is intended for use in the IM-192-WDQ and IM-151/WDQ Radiacmeters. When installed in these systems, the pump pulls air through filter paper which traps airborne dust particles. The systems then measure the radioactivity of these particles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification with number of slash sheet chosen for procurement.
(b) Packaging and packing requirements (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisiiton which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD For 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulation (DAR) 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5	Report, Acceptance Test	UDI-T-22741A
4.3	First Article Inspection Report	DI-T-4902
4.5.7	Procedures, Reliability Tests	DI-R-7035
4.5.7	Reports, Reliability Test and Demonstration	DI-R-7034

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DOD 5000.19L., Vol II, AMSL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The

Source: https://assist.dla.mil -- Downloaded: 202640-20T19:15Z
Check the source to verify that this is the current version before use.

contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.3.1 Conditions for waiver of first article requirement. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested in accordance with this specification by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement. Such approval will not be extended more than three years since the last test.

6.4 Test equipment required. A recommended list of test equipment is provided for information in Appendix B.

6.5 Warranty. A warrenty clause will be invoked. A registration card and notice of warranty will be included with each pump.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - EC

(Proj. No. 4310-N096)

Source: https://assist.dla.mil -- Downloaded: 20261O-20T19:15Z
Check the source to verify that this is the current version before use.

APPENDIX A

RADIACMETER IM-192/WDQ DRY AIR VACUUM
PUMP TEST PROCEDURES

10. SCOPE

10.1 Scope. This appendix details the specific test procedures for the Radiacmeter IM-192/WDQ Dry Air Vacuum Pump.

20. TEST PROCEDURES

20.1 Preliminary operation. The pump shall be operated for at least ten hours prior to the start of the test measurements. Inlet vacuum should be set at 8 inches of mercury and output pressure at 2 psig. A temperature/humidity chamber shall be used to provide temperature/humidity conditions when required.

NOTE

Although two separate diagrams are used in these procedures to illustrate the test fixture configuration, in actual practice only one test fixture suffices for all of the tests. Valves and guages are arranged and utilized to select the appropriate test configuration.

30. OPERATING PERFORMANCE TEST

30.1 Test configuration. A suitable configuration for measuring the performance parameters of the pump is connected as shown in Figure 3. This set-up is used to measure the inlet vacuum, outlet pressure, air volume flow rate, and input power test. The ambient condition shall be STP as defined in Section 3 unless otherwise specified.

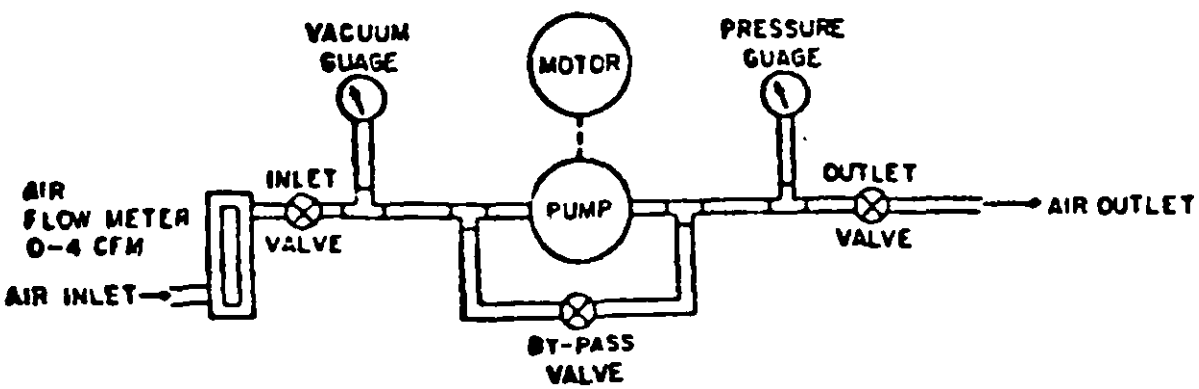


FIGURE 3. Test set-up for performance parameter measurements of dry air vacuum pump of radiacmeter IM-192/WDQ.

30.2 Inlet vacuum. With the pump running at 1420 rpm, the inlet valve and the pump by-pass valve closed, and the outlet valve open; the inlet vacuum on the vacuum guage shall be at least 15 inches of mercury. (See Figure 2)

Source: https://assist.dla.mil -- Downloaded: 202610-20T19:15Z
Check the source to verify that this is the current version before use.

NOTE

Mercury manometers, equipment containing mercury, or mercury compounds shall not be used in any manner.

30.3 Outlet pressure. With the pump running at 1420 rpm, the outlet valve and the by-pass valve closed, and the inlet valve open; the outlet pressure on the pressure guage shall be at least 15 psig. (See Figure 3).

NOTE

Do not operate the pump at the high outlet pressure setting for more than 10 minutes. This procedure will place a high stress on the pump.

30.4 Air volume flow rate. With the pump rotating at 1420 rpm, the pump by-pass valve closed and the inlet and outlet valves open, the maximum air volume flow rate on the flowmeter shall be at least 3.5 cfm. Adjust the inlet and outlet valves to an inlet vacuum of 8 inches of mercury and an outlet pressure of 2 psig. The air volume flow rate shall be at least 2.1 cfm. (See Figure 3)

30.5 Efficiency test.

30.5.1 The efficiency test is conducted by measuring the input power to the electric motor which drives the vacuum pump, while observing the output readings of the pump.

30.5.2 A wattmeter (1500 watt range) is inserted in the power line feeding motor. The test configuration shown in Figure 3 applies. With the pump rotating at 1420 rpm and the pump by-pass valve closed, adjust the inlet and outlet valves to obtain an inlet vacuum of 8 inches of mercury and an outlet pressure of 2 psig. The input power on the wattmeter shall not exceed 600 watts.

30.5.3 Monitor the pump case temperature. The sensing element of the thermometer should be placed in firm contact with the top center of the pump housing in the valley between the fins. Do not use a mercury filled thermometer. An all metal or alcohol thermometer or thermocouple type thermometer is suitable for this purpose. The pump case temperature rise shall not exceed 50°C above the ambient room temperature, while operating under the test coditions of Appendix A, 30.5.2.

30.6 Leakage test. The leakage test is performed to determine the leakage rate of the dry air vacuum pump. This test will provide a check on the quality of the shaft installation. The connection schematic for the leakage test is shown in Figure 4.

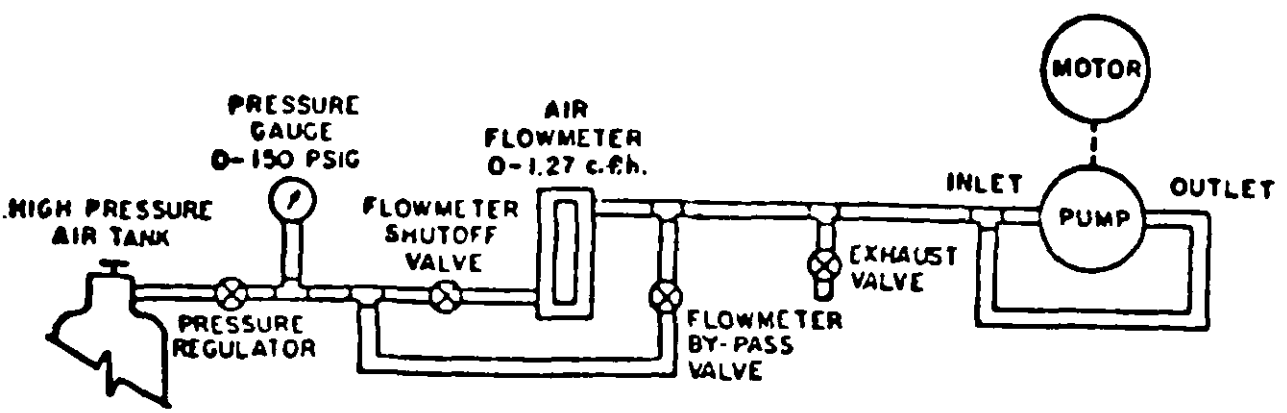


FIGURE 4. Connection schematic for pressure test of dry air vacuum pump of radiacmeter IM-192/WDQ.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T19:15Z
Check the source to verify that this is the current version before use.

NOTE

Extreme caution should be used when performing this test due to the high air pressure involved with a high pressure air tank connected to the system. First check for leaks in the fittings at low pressure.

30.6.1 The leakage rate will be monitored for 1 minute while maintaining the system pressure at 150 psig as follows:

(a) Close the flowmeter shutoff valve and open the flowmeter by-pass valve.
(b) Start the vacuum pump operating.
(c) Close the exhaust valve.
(d) Close the regulator valve.
(e) Open the high pressure bottle valve.
(f) Open the regulator valve and slowly increase the regulator pressure to 150 psig.
(g) When the system stops filling, close the flowmeter by-pass valve and open the flowmeter valve.
(h) Read the system pressure. It shall be 150 psig.
(i) Read the leakage rate on the flowmeter. Leakage rate should not exceed 1.0 cfh. Do not operate the pump with the closed system for more than 10 minutes; the pump will overheat if there is no significant air flow through the system.

30.6.2 De-pressurize the system as follows:

(a) Close the high pressure bottle valve.
(b) Shut off electrical power to the pump motor.
(c) Open the flowmeter by-pass valve.
(d) Close the flowmeter shutoff valve.
(e) Slowly open the exhaust valve.
(f) Ensure that all guages read zero.
(g) Close the pressure regulator valve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:15Z
Check the source to verify that this is the current version before use.

APPENDIX B

EQUIPMENT RECOMMENDED FOR TEST FIXTURE

10. SCOPE.

10.1 Scope. This appendix details the materials required for conducting the performance parameter measurements and pressure tests.

20. PERFORMANCE PARAMETER MEASUREMENTS

20.1 Materials required:

Item	Quantity	Description	Part No.
1	(3)	Valve, 3/4 inch	
2	(1)	Guage, pressure, 0-30 psig	
3	(1)	Guage vacuum, 0-30 inches of mercury	
4	(1)	Flowmeter, 0-05.5 cfm calibrated at STP, 70 degrees Fahrenheit 14.7 psia	Brooks Instrument Division, Emerson Electric Co. Model 1110 or equivalent
5	(1)	Tube, Copper, 3/4 inch o.d., 7.62 meter (25 ft) roll.	
6	as required	Tube fittings	
7	(1)	Motor, Alternating	9C 6105-00-799-3542
8	(1)	Belt, V	9C 3030-00-528-4279
9	(1)	Pulley, Browning	AK34 or equivalent
10	(1)	Pulley, Browning	AK44 or equivalent
11	(1)	Wattmeter 0-1500 watts	Simpson model 1379 or equivalent
12	(1)	Thermometer, 0-100 degrees Centigrade	
13	(1)	Probe, surface temperature	Fisher Scientific Co. Cat. No. 15-176-35 or equivalent

Source: https://assist.dla.mil -- Downloaded: 202610-20T19:15Z
Check the source to verify that this is the current version before use.

30. PRESSURE TESTS

30.1 Materials required:

Item	Quantity	Description	Part No.
1	(1)	Compressed Air, Technical	6380-00-169-0774
2	(1)	Tank, 200 cubic feet (not required if trade-in is available)	8120-00-151-9745
3	(1)	Regulator, Pressure, Air 0-150 psig	6680-00-171-9816
4	(1)	Flowmeter Model 1355 with R-2-15-AAA tube, or equivalent, 0-1.27 cubic feet per hour at 70 degrees Fahrenheit, 150 psic	Brooks Instrument Division, Emerson Electric Company Hatfield, PA 19440 or equivalent
5	(4)	Valve	
6	(1)	Tube, Copper 3/4 inch o.d., 7.62 meters (25 ft) roll	
7	as required	Tube fittings	
8	(1)	Motor, Alternating	9C 6105-00-799-3542
9	(1)	Belt, V	9C 3030-00-528-4279
10	(1)	Pulley, Browning	AK34, or equivalent
11	(1)	Pulley, Browning	AK44, or equivalent

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:15Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-28824A(EC)	Pump, Vacuum, Positive Displacement, Air Rotary Vane

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-20T19:15Z
Check the source to verify that this is the current version before use.